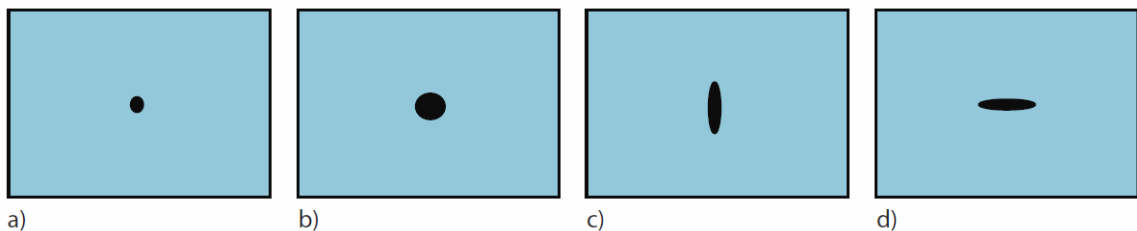


Submission details

- This is a **LONG** assignment. Please don't wait with it for the last minute.
- This assignment should be done in singles or in pairs.
- The topics of this assignment is The Pinhole Camera, Camera Calibration, Single View Metrology
- The submission date is **30/07/2024**. Please pay attention to the late submission policy.
- Coding should be done in Python.
- You are not required to use any specific function or library, unless stated otherwise. If in doubt, please contact me via email or Moodle forum.
- For submission, package up your code as a zip file. Include your written answers as a pdf file named writeup.pdf. Include graphs and images in your writeup, and please label them so we know which figures go with which sub-problem. Alternatively, write everything inside your python notebook.
- Name the zip file with ID number(s). Submit your zip file in Moodle.
- If you have any questions about this assignment, please contact me.

Task 1: The Pinhole Camera

1. Consider a pinhole camera in which we change the geometry of the aperture as shown in the following image:



Consider a basic scene situated in front of the pinhole camera. Illustrate the images that would be taken by the pinhole camera for each of the four aperture shapes. Explain what leads you to your answers. Note that this question is qualitative. It is not important to take into account the exact size of the aperture.

2. Explain the problems affecting an image taken by a pinhole camera when the diameter of the aperture is too small, and when the diameter of the aperture is too large.

Task 2: Camera Calibration

Here the goal is to compute the 3×4 camera matrix P describing a pinhole camera, given the coordinates of 10 world points and their corresponding image projections. Then you will decompose P into the intrinsic and extrinsic parameters. You should write a simple Python script that works through the stages below, printing out the important terms.

Use the provided ASCII files, world.txt and image.txt. The first file contains the (X,Y,Z) values of 10 world points, and the second file contains the (x,y) projections of those 10 points.

Add to your writeup significant values resulting from the stages below.

1. Find the 3x4 matrix P that projects the world points $\underline{X}$ to the 10 image points $\underline{x}$. This should be done in the following steps:

- Since P is a homogeneous matrix, the world and image points (which are 3-D and 2-D respectively), need to be converted into homogeneous points by concatenating a 1 to each of them (thus becoming 4-D and 3-D respectively).
- We now note that $\underline{x} \times P\underline{X} = \underline{0}$, irrespective of the scale ambiguity. This allows us to setup a series of linear equations of the form:

$$\begin{bmatrix} \underline{0}^T & -w_i \underline{X}_i^T & y_i \underline{X}_i^T \\ w_i \underline{X}_i^T & \underline{0}^T & -x_i \underline{X}_i^T \\ -y_i \underline{X}_i^T & x_i \underline{X}_i^T & \underline{0}^T \end{bmatrix} \begin{pmatrix} P_1^T \\ P_2^T \\ P_3^T \end{pmatrix} = \underline{0}$$

for each correspondence $\underline{x}_i \leftrightarrow \underline{X}_i$, where $\underline{x}_i = (x_i, y_i, w_i)^T$, w_i being the homogeneous coordinate, and P_j is the j^{th} row of P . But since the 3rd row is a linear combination of the first two, we need only consider the first two rows for each correspondence. Thus, you should form a 20 by 12 matrix A , each of the 10 correspondences contributing two rows. This yields $A\underline{p} = \underline{0}$, $\underline{p}$ being the vector containing the entries of matrix P .

- To solve for $\underline{p}$, we need to impose an extra constraint to avoid the trivial solution $\underline{p} = \underline{0}$. One simple one is to use $\|\underline{p}\|_2 = 1$. This constraint is implicitly imposed when we compute the SVD of A . The value of $\underline{p}$ that minimizes $A\underline{p}$ subject to $\|\underline{p}\|_2 = 1$ is given by the eigenvector corresponding to the smallest singular value of A . To find this, compute the SVD of A , picking this eigenvector and reshaping it into a 3x4 matrix P .
- Verify your answer by re-projecting the points $\underline{X}$ and checking that they are close to $\underline{x}$.

Now that we have P , we can compute the world coordinates of the projection center of the camera $\underline{C}$. Note that $P\underline{C} = \underline{0}$, thus $\underline{C}$ lies in the null space of P , which can again be found using SVD. Compute the SVD of P and pick the vector corresponding to this null-space. Finally, convert it back to homogeneous coordinates to yield the (X,Y,Z) coordinates.

Task 3: Single View Metrology

1. A chess board of 1x1 meters is aligned such that its lines are parallel to the axes X and Z , and one of its corners is at the origin of the axes. Each of the corners between the board's checkers is located in the position X_{ij} that fits a 2D integer index (i, j) , $0 \leq i, j \leq 8$. The origin of the axes fits the index $(0,0)$: $X_{00} = (0,0,0)^T$. An ideal camera with a focal length f is located at $X_c = (0,1,0)^T$ (one meter above the origin), and is directed at an angle of θ relative to the Z axis such that the camera's main axis is parallel to the board, and one of the axes of the image plane is parallel to the Y axis. Assume that the size of the focal plane and the image in it is unlimited.

- a. Calculate the locations of the corners of the board's checkers in the 3D world as a function of (i, j) .
 - b. Calculate the locations of the corners of the board's checkers in the image plane as a function of (i, j) and θ .
 - c. Calculate the location of the vanishing line belonging to the plane of the chess board.
2. Properties of the orthographic projection.
- a. Suppose we have a line given by a point $\mathbf{p}$ and a direction vector $\mathbf{v}$ so that the equation for the line is $\mathbf{l}(t) = \mathbf{p} + t\mathbf{v}$. A parallel line would be any line that has the same direction $\mathbf{v}$. Show that under orthographic projection, parallel lines project to parallel lines.
 - b. Given a line from $\mathbf{a}$ to $\mathbf{b}$, and any point $\mathbf{c}$ along this line, show that the quantity

$$\frac{|\mathbf{c} - \mathbf{a}|}{|\mathbf{b} - \mathbf{c}|}$$

is invariant under orthographic projection (Hint: $\mathbf{c}$ can be expressed as $\mathbf{a} + t(\mathbf{b} - \mathbf{a})$, where $0 \leq t \leq 1$).

- c. Why do the above properties not hold for perspective projection? Explain briefly.
3. Let the vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, $\mathbf{d}$ correspond to the corners of a square mapped under perspective projection, where the homogeneous coordinates of the vectors are

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix}, \mathbf{c} = \begin{bmatrix} 15 \\ 12 \\ 3 \end{bmatrix}, \mathbf{d} = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$$

Calculate the vanishing point for each pair of parallel lines. Express your answer in homogeneous coordinates (Hint: Try plotting the points on a plane).

4. Bonus: Prove that under perspective projection, all the vanishing points that are received from lines parallel to a certain plane (which are not perpendicular to the image plane) lie on straight line.

Task 4: Object Detection in Images

In this task you will build a system for specific object detection. We're providing you with 3 datasets and it's ok to have a solution that's good only for this data. If you'd rather make it broader that's also fine. Creativity is a key factor in this assignment. Thinking out-of-the-box could lead to better performance. A possible idea is to search for relevant papers and follow the footsteps of one that you like. The detection performance will have some part in the grading (but not a big one).

1. Propose a method for finding a given template image in a set of query images. Any method you propose is fine, as long as you explain and motivate it clearly. Before you code it and test it, please write the following:
 - a. The motivation to the method you propose.
 - b. The guiding principles of your approach.
 - c. In what cases do you expect it to succeed or fail?
 - d. What are the advantages and disadvantages of your approach?

2. Implement the method and apply it to the 3 provided datasets. To evaluate the quality of the results, draw on each query image green bounding-boxes of the detected templates.
3. Based on your experimental results, revisit item 1 and assess your predictions. Associate examples from your results with items on your list of pros & cons. Explain the success/failure cases you got.
4. Suggest a modification to your method that will improve at least one of the failure modes, and implement it. Did it work? Explain why.